

ATP FIX 4.4 Quote API Specification
v1.5

DOCUMENT HISTORY

Version	Date	Changes
v1.5.1	20250411	1. Fix value of ' Limit High Price ' and ' Limit Low Price ' of ' MDEntryType '
v1.5	20241022	1. Update description of ' MDEntryPx ' and ' MDEntrySize ' to include blank value
v1.4	20240603	2. Optimized the document
v1.3	20230419	1. Update message type Market Data Request (35=V) 2. Delete message type Mass Quote (35=i) 3. Add message type Market Data Snapshot Full Refresh (35=W) 4. Add message type Market Data Snapshot Incremental Refresh (35=X)
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1. Introduction

1.1. Disclaimer

This confidential document is the property of ATP and neither the document nor its contents may be disclosed to a third party, nor may it be copied, without ATP's prior written consent.

ATP endeavors to ensure that the data and other material in this publication are correct and complete but does not accept liability for any error herein or omissions here from. The development of ATP's products and services is continuous and published information may not be up to date.

1.2. Scope Of This Document

This document is intended to serve software developers as an implementation guide for the ATP FIX API.

1.3. FIX Version

ATP supports FIX version 4.4, but with custom message types and fields (and more custom message types and fields may be added in the future). For further information about this version please refer to the specifications published by the FIX Protocol Organization under <https://www.fixtrading.org/standards/fix-4-4>

2. Connectivity

2.1. Connection Type

Connection to ATP's FIX engine is available via TCP/IP (**without** SSL/TLS).

2.2. Hours Of Operations

Connectivity to ATP's FIX engine is available on following time (America/New_York):

- SUN 18:00 - FRI 17:30

2.3. Sequence Number Reset

ATP FIX server will reset the sequence number each time once the client connection is disconnected. Client connections have to be configured to reset sequence numbers on logon (141=Y).

2.4. Security And Authentication

Once the connection to ATP FIX server is established, client should authenticate against the server with username and password added to the Logon(35=A) message.

3. Message Overview

Each FIX message consists of a stream of <tag>=<value> fields. A fix message consists of 3 elements, a Header, a Body and a Trailer. All fields in a FIX message are terminated by ASCII SOH (#001) character.

- **Header:** Included in each message, it mainly identifies the message type, length and routing.
- **Body:** It defines the content of the actual business level data.
- **Trailer:** Included in each message with checksum value.

3.1. Supported Message Types (35=?)

Messages that are not in this list will be rejected.

Message Type	Message Name	Message Direction
A	Logon	Client <-> ATP
0	Heartbeat	Client <-> ATP
1	Test Request	Client <-> ATP
2	Resend Request	Client <-> ATP
3	Reject	ATP -> Client
4	Sequence Reset	Client <-> ATP
5	Logout	Client <-> ATP
V	Market Data Request	Client -> ATP
Y	Market Data Request Reject	ATP -> Client
W	Market Data Snapshot Full Refresh	ATP -> Client
X	Market Data Snapshot Incremental Refresh	ATP -> Client

3.2. Data Types

Each field within a FIX message has an associated data type allowing for data validation.

Type	Comments
Int	Sequence of digits without commas or decimals and optional sign character (ASCII characters "-" and "0" - "9"). The sign character utilizes one byte (i.e. positive int is "99999" while negative int is "-99999").
Float	Sequence of digits with optional decimal point and sign character (ASCII characters "-", "0" - "9" and "."); the absence of the decimal point within the string will be interpreted as the float representation of an integer value. All float fields must accommodate up to 15 digits. The number of decimal places used should be a factor of business/market needs and mutual agreement between counterparties
Char	Single character value, can include any alphanumeric character or punctuation except the delimiter. All char fields are case sensitive (i.e. m - M)
String	Alpha-numeric free format strings can include any character or punctuation except the delimiter. All char fields are case sensitive (i.e. morstatt - Morstatt)
Boolean	A char field (see definition of "Char" above) containing one of two values: 'Y' = True/Yes

	'N' = False/No
Timestamp	String field (see definition of "String" above) representing the Time/date combination in UTC (Universal Time Coordinated, also known as "GMT") in YYYYMMDD-HH:MM:SS.sss (milliseconds) format, colons, dash, and period required.
Price	Float field (see definition of "Float" above) representing a price.
SeqNum	Int field (see definition of "Int" above) representing a message sequence number. Value must be positive.
Qty	Int field (see definition of "Int" above) capable of storing quantities of derivatives, such as futures and options, denominated in whole units for trading purposes.
NumInGroup	Int field (see definition of "Int" above) representing the number of entries in a repeating group. Value must be positive.

3.3. General Message Header

All FIX messages use the standard message header, as specified by the FIX 4.4 specification, below describes the tags associated with the message header. Any message that does not have these tags in the correct order will be rejected.

Tag	Field Name	Req'd	Data Type	Comments
8	BeginString	Y	String	Identifies beginning of new message and protocol version (always first field in message)
9	BodyLength	Y	Int	Message length (in bytes) forward to the CheckSum field (always second field in message)
35	MsgType	Y	String	Defines message type (always 3rd tag in message)
49	SenderCompID	Y	String	Assigned value used to identify the Client sending messages (will be provided by ATP)
56	TargetCompID	Y	String	Assigned value used to identify receiving party (will be provided by ATP)
34	MsgSeqNum	Y	SeqNum	Integer message sequence number
52	SendingTime	Y	Timestamp	Message transmission time in UTC/GMT

3.4. General Message Trailer

Each message is terminated by a standard trailer. The trailer is used to segregate messages and contains the three-digits characters representation of a checksum value:

Tag	Field Name	Req'd	Data Type	Comments
10	Checksum	Y	Int	Three digits characters representing the checksum value of the message

4. Session Messages

4.1. Logon(35=A) (Client <-> ATP)

After establishing your TCP connection with the ATP FIX server, a Logon(35=5) must be the first message that the FIX client sends.

Possible response message:

- Logon(35=A)
- Logout(35=5)

Tag	Field Name	Req'd	Data Type	Comments
	Standard Header	Y		MsgType=A
98	EncryptMethod	Y	Int	Use of Encryption, set to "0"
108	HeartBtInt	Y	Int	Heart beat interval in seconds
141	ResetSeqNumFlag	Y	Boolean	Always set to "Y"
553	Username	Y	String	Username (provided by Client)
554	Password	Y	String	Password (provided by Client)
	Standard Trailer	Y		

Logon Example

Client -> ATP: 8=FIX.4.4 9=101 35=A 49=CLIENT01 56=ATPFIX 34=1 52=20240604-02:09:01.575 98=0 108=10 141=Y 553=user01 554=password01 10=233

ATP -> Client: 8=FIX.4.4 9=75 35=A 49=ATPFIX 56=CLIENT01 34=1 52=20240604-02:09:01.824 98=0 108=10 141=Y 10=021

4.2. Heartbeat(35=0) (Client <-> ATP)

The Heartbeat monitor the status of the communication link. It is sent automatically based on the interval set in the logon message.

Tag	Field Name	Req'd	Data Type	Comments
	Standard Header	Y		MsgType=0
112	TestReqID	N	String	Req'd when the heartbeat is the result of a Test Request(35=1) message
	Standard Trailer	Y		

Heartbeat Example

Client -> ATP: 8=FIX.4.4 9=57 35=0 49=CLIENT01 56=ATPFIX 34=3 52=20210304-06:35:46.673 10=202

4.3. Test Request(35=1) (Client <-> ATP)

The Test Request is used to force a heartbeat in order to verify connectivity status and sequence number generation.

Possible response message:

- Heartbeat(35=0)

Tag	Field Name	Req'd	Data Type	Comments
	Standard Header	Y		MsgType=1

112	TestReqID	Y	String	A unique identifier for this test message
	Standard Trailer	Y		

Test Request Example

Client -> ATP: 8=FIX.4.4 9=67 35=1 49=CLIENT01 56=ATPFIX 34=2 52=20210304-06:39:43.570 112=reqID 10=111

4.4. Resend Request (35=2) (Client <-> ATP)

In the event that a sequence number gap is detected, the Resend Request message notifies the opposite party to resend message.

Tag	Field Name	Req'd	Data Type	Comments
	Standard Header	Y		MsgType=2
7	BeginSeqNo	Y	Int	Message sequence number of first message in range to be resent
16	EndSeqNo	Y	Int	Message sequence number of last message in range to be resent. If request is for a single message 'BeginSeqNo' = 'EndSeqNo'. If request is for all messages subsequent to a particular message, 'EndSeqNo' = 0 (representing infinity)
	Standard Trailer	Y		

Resend Request Example

Client -> ATP: 8=FIX.4.4 9=66 35=2 49=CLIENT01 56=ATPFIX 34=8 52=20210304-07:43:46.588 7=5 16=0 10=085

4.5. Reject (35=3) (ATP -> Client)

ATP FIX server uses this message to respond to a message that could not be processed. Rejections for unsupported tags, invalid message format, or field restrictions are also identified. Where applicable, the reason for the reject is included in tag 58. This session level reject message can be sent in response to any message sent from the FIX client to the ATP FIX server.

Tag	Field Name	Req'd	Data Type	Comments
	Standard Header	Y		MsgType=3
45	RefSeqNum	Y	Int	MsgSeqNum of rejected message
371	RefTagID	N	Int	The tag number of the FIX field being referenced
372	RefMsgType	N	String	The MsgType of the FIX message being referenced
373	SessionRejectReason	N	Int	Code to identify reason for a session-level Reject(35=3) message. Valid values: 1 = Required tag missing 4 = Tag specified without a value 5 = Value is out of range for this tag 11 = Invalid MsgType 99 = Other
58	Text	N	String	Message to explain reason for rejection
	Standard Trailer	Y		

Reject Request Example

ATP -> Client: 8=FIX.4.4 9=133 35=3 49=ATPFIX 56=CLIENT01 34=2 52=20210304-07:57:05.640 45=2 58=Invalid Resend Request: BeginSeqNo (5) is greater than expected (1) 10=118

4.6. Sequence Reset (35=4) (Client <-> ATP)

In the event that either system detects an error in sequence numbers, this message is sent to rectify the situation.

Tag	Field Name	Req'd	Data Type	Comments
	Standard Header	Y		MsgType=4
123	GapFillFlag	N	Boolean	Indicates that the Sequence Reset(35=4) message is replacing administrative or application messages that will not be resent. Valid values: Y = Gap Fill message, MsgSeqNum field valid N = Sequence Reset, ignore MsgSeqNum
36	NewSeqNo	Y	Int	New sequence number
	Standard Trailer	Y		

Sequence Reset Example:

ATP -> Client: 8=FIX.4.4 9=100 35=4 49=ATPFI 56=CLIENT01 34=5 43=Y 52=20210305-10:48:34.613 122=20210305-10:48:34.613 123=Y 36=15 10=033

4.7. Logout (35=5) (Client <-> ATP)

The Logout(35=5) message is the last message sent on the connection. Either the ATP FIX server or a FIX client can send the Logout(35=5) message.

Tag	Field Name	Req'd	Data Type	Comments
	Standard Header	Y		MsgType=5
58	Text	N	String	Reason for logout
	Standard Trailer	Y		

Logout Example

Client -> ATP: 8=FIX.4.4 9=57 35=5 49=CLIENT01 56=ATPFI 34=9 52=20210304-06:43:13.746 10=207

ATP -> Client: 8=FIX.4.4 9=84 35=5 49=ATPFI 56=CLIENT01 34=11 52=20210304-06:43:13.747 58=Logout acknowledgement 10=118

5. Application Messages

5.1. Market Data Request (35=V) (Client -> ATP)

The FIX client sends this message to subscribe/unsubscribe to market data for a specific symbol from the ATP FIX server.

Possible response message:

- Reject(35=3)
- Market Data Request Reject(35=Y)
- Market Data Snapshot Full Refresh(35=W)
- Market Data Snapshot Incremental Refresh(35=X)

Note:

- When the correct subscription is requested, the Market Data Snapshot Full Refresh(35=W) message containing all the requested types of Market Data Entries will be received in response, followed by possible Market Data Snapshot Incremental Refresh(35=X) message

Tag	Field Name	Req'd	Data Type	Comments
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	Standard Header		Y		MsgType=V
262	MDReqID		Y	String	Must be Unique, or the ID of previous Market Data Request (35=V) to disable if 'SubscriptionRequestType' = 2 (Disable previous snapshot plus update request). The accepted length is limited to a maximum of 3 characters
263	SubscriptionRequestType		Y	Char	1 = Snapshot plus updates (subscribe) 2 = Disable previous snapshot plus update request (unsubscribe)
264	MarketDepth		C	Int	The level of market depth (non-negative number). Required if 'SubscriptionRequestType' = 1 (Snapshot plus updates)
265	MDUpdateType		C	Int	Specifies the type of Market Data Update. Required if 'SubscriptionRequestType' = 1 (Snapshot plus updates). Valid values: 1 = Incremental Refresh
146	NoRelatedSym		Y	NumInGroup	Always set to "1"
→	55	Symbol	Y	String	Name of the symbol
267	NoMdEntryTypes		C	NumInGroup	Number of 'MDEntryType' fields requested. Required if 'SubscriptionRequestType' = 1 (Snapshot plus updates)
→	269	MDEntryType	C	Char	List of all the types of Market Data Entries that the firm requesting the Market Data is interested in receiving. Required if 'SubscriptionRequestType' = 1 (Snapshot plus updates). Valid values: 0 = Bid(not available if MarketDepth=0) 1 = Offer(not available if MarketDepth=0) 2 = Trade 4 = Open Price 5 = Pre-Close Price 6 = Pre-Settlement Price 7 = High Price 8 = Low Price B = Total Volume C = Open Interest D = Limit High Price E = Limit Low Price
	Standard Trailer		Y		

Market Data Request Example**Subscribe**

Client -> ATP: 8=FIX.4.4 9=180 35=V 49=CLIENT01 56=ATPFIX 34=2 52=20230419-08:58:05.144 262=re1 263=1 264=1 265=1 267=12 269=0 269=1 269=2 269=4 269=5 269=6 269=7 269=8 269=B 269=C 269=D 269=E 146=1 55=CU3M-LME 10=089

Unsubscribe

Client -> ATP: 8=FIX.4.4 9=89 35=V 49=CLIENT01 56=ATPFIX 34=8 52=20230419-08:59:05.148 262=re1 263=2 146=1 55=CU3M-LME 10=195

5.2. Market Data Request Reject (35=Y) (ATP -> Client)

The message serves as a notification of rejection for the market data request.

ATP FIX server sends this message in response to:

- Market Data Request(35=V)

Tag	Field Name	Req'd	Data Type	Comments
	Standard Header	Y		MsgType=Y
262	MDReqID	Y	String	ID of the market data request
58	Text	N	String	Reason for market data request being rejected
	Standard Trailer	Y		

Market Data Request Reject Example

ATP -> Client: 8=FIX.4.4 9=102 35=Y 49=ATPFIX 56=CLIENT01 34=2 52=20210305-10:51:21.955 262=re1 58=Symbol was not subscribed before. 10=243

5.3. Market Data Snapshot Full Refresh (35=W) (ATP -> Client)

The message serves as a notification of the market data snapshot for the subscribed symbol.

ATP FIX server sends this message in response to:

- Market Data Request(35=V)

Note:

- Market Data Snapshot Full Refresh(35=W) message will be published in response to subscription
- The message contains the initial snapshot (the initial value of all the requested types of Market Data Entries, such as bid, offer, high price, low price, and so on). If the snapshot changes latter, these changes will be published via Market Data Snapshot Incremental Refresh(35=X) message

Tag	Field Name		Req'd	Data Type	Comments
	Standard Header		Y		MsgType=W
262	MDReqID		Y	String	Unique identifier for Market Data Request (35=V). Use this value to identify the symbol to which the data in this message refers to
268	NoMDEntries		Y	NumInGroup	Number of entries following
→	269	MDEntryType	Y	Char	Type of Market Data Entry. Must be the first field in this repeating group. Valid values: 0 = Bid(not available if MarketDepth=0) 1 = Offer(not available if MarketDepth=0) 2 = Trade 4 = Open Price 5 = Pre-Close Price 6 = Pre-Settlement Price 7 = High Price 8 = Low Price B = Total Volume C = Open Interest D = Limit High Price E = Limit Low Price
→	270	MDEntryPx	C	Price	Price of the Market Data Entry. This field may be blank if no data exists for the corresponding Market Data Entry. Conditionally required if 'MDEntryType' is not 'B'(Trade Volume), or 'C'(Open Interest)
→	271	MDEntrySize	C	Qty	Quantity or volume represented by the Market Data Entry. This field may be blank if no data exists for the corresponding Market Data Entry.

					Conditional required if 'MDEntryType' is '0'(Bid), '1'(Offer), '2'(Trade), 'B'(Trade Volume), or 'C'(Open Interest)
→	290	MDEntryPositionNo	C	Int	Display position of a bid or offer, numbered from most competitive to least competitive, per market side, beginning with 1. Conditional required if 'MDEntryType' is '0'(Bid), '1'(Offer)
	Standard Trailer		Y		

Market Data Snapshot Full Refresh Example

ATP -> Client: 8=FIX.4.4 9=304 35=W 49=ATPFIX 56=CLIENT01 34=2 52=20230419-08:58:05.135 262=re1 268=12 269=2 270=8899 271=1 269=B 271=5651 269=C 271=0 269=D 270=10364 271=10000 269=E 270=7661 271=1 269=4 270=9004 269=5 270=9002.5 269=7 270=9011.5 269=8 270=8876 269=6 270=9012.5 269=0 270=8898 271=1 290=1 269=1 270=8900.5 271=1 290=1 10=121

5.4. Market Data Snapshot Incremental Refresh (35=X) (ATP -> Client)

The message serves as a notification of the incremental market data update for the subscribed symbol.

ATP FIX server sends this message in response to:

- Market Data Request(35=V)

Note:

- Before Market Data Snapshot Incremental Refresh(35=X) message can be published, there must be a Market Data Snapshot Full Refresh(35=W) message that has been published to indicate the initial snapshot of the requested symbol
- Market Data Snapshot Incremental Refresh(35=X) message is used for snapshot incremental updates, so the number of messages depends on the changes of the snapshot. If the snapshot of the symbol does not change after initial snapshot which is published, the Market Data Snapshot Incremental Refresh(35=X) message will not be published; if the snapshot of the symbol changes continuously, the Market Data Snapshot Incremental Refresh(35=X) message will be published multiple times
- For the bid/offer orderbook, when there are differences between the latest orderbook and the published orderbook, these differences will be aggregated and published via Market Data Snapshot Incremental Refresh(35=X) message. Clients can reconstruct the orderbook from these messages

Tag	Field Name		Req'd	Data Type	Comments
	Standard Header		Y		MsgType=X
262	MDReqID		Y	String	Unique identifier for Market Data Request (35=V). Use this value to identify the symbol to which the data in this message refers to
268	NoMDEntries		Y	NumInGroup	Number of entries following
→	279	MDUpdateAction	Y	Char	Market Data update action. Must be first field in this repeating group. Valid values: 1 = Change
→	269	MDEntryType	Y	Char	Type of Market Data Entry. Valid values: 0 = Bid(not available if MarketDepth=0) 1 = Offer(not available if MarketDepth=0) 2 = Trade 4 = Open Price 5 = Pre-Close Price 6 = Pre-Settlement Price 7 = High Price 8 = Low Price B = Total Volume C = Open Interest

					D = Limit High Price E = Limit Low Price
→	270	MDEntryPx	C	Price	Price of the Market Data Entry. This field may be blank if no data exists for the corresponding Market Data Entry. Conditionally required if 'MDEntryType' is not 'B'(Trade Volume), or 'C'(Open Interest)
→	271	MDEntrySize	C	Qty	Quantity or volume represented by the Market Data Entry. This field may be blank if no data exists for the corresponding Market Data Entry. Conditional required if 'MDEntryType' is '0'(Bid), '1'(Offer), '2'(Trade), 'B'(Trade Volume), or 'C'(Open Interest)
→	290	MDEntryPositionNo	C	Int	Display position of a bid or offer, numbered from most competitive to least competitive, per market side, beginning with 1. Conditional required if 'MDEntryType' is '0'(Bid), '1'(Offer)
	Standard Trailer		Y		

Market Data Snapshot Incremental Refresh Example

ATP -> Client: 8=FIX.4.4 9=140 35=X 49=ATPFIX 56=CLIENT01 34=24 52=20230419-08:58:36.474 262=re1 268=2 279=1 269=0 270=8897.5 271=2 290=1 279=1 269=1 270=8900 271=1 290=1 10=118

ATP -> Client: 8=FIX.4.4 9=120 35=X 49=ATPFIX 56=CLIENT01 34=30 52=20230419-08:58:49.269 262=re1 268=2 279=1 269=2 270=8900 271=1 279=1 269=B 271=5652 10=252

Full Market Data Example

Client -> ATP: 8=FIX.4.4 9=180 35=V 49=CLIENT01 56=ATPFIX 34=2 52=20230419-08:58:05.144 262=re1 263=1 264=1 265=1 267=12 269=0 269=1 269=2 269=4 269=5 269=6 269=7 269=8 269=B 269=C 269=D 269=E 146=1 55=CU3M-LME 10=089

ATP -> Client: 8=FIX.4.4 9=304 35=W 49=ATPFIX 56=CLIENT01 34=2 52=20230419-08:58:05.135 262=re1 268=12 269=2 270=8899 271=1 269=B 271=5651 269=C 271=0 269=D 270=10364 271=10000 269=E 270=7661 271=1 269=4 270=9004 269=5 270=9002.5 269=7 270=9011.5 269=8 270=8876 269=6 270=9012.5 269=0 270=8898 271=1 290=1 269=1 270=8900.5 271=1 290=1 10=121

ATP -> Client: 8=FIX.4.4 9=140 35=X 49=ATPFIX 56=CLIENT01 34=24 52=20230419-08:58:36.474 262=re1 268=2 279=1 269=0 270=8897.5 271=2 290=1 279=1 269=1 270=8900 271=1 290=1 10=118

ATP -> Client: 8=FIX.4.4 9=120 35=X 49=ATPFIX 56=CLIENT01 34=30 52=20230419-08:58:49.269 262=re1 268=2 279=1 269=2 270=8900 271=1 279=1 269=B 271=5652 10=252

Client -> ATP: 8=FIX.4.4 9=89 35=V 49=CLIENT01 56=ATPFIX 34=8 52=20230419-08:59:05.148 262=re1 263=2 146=1 55=CU3M-LME 10=195

6. Information Needed From ATP

Please see below the information required in order for Client to connect and perform UAT conformance testing:

Field Name	Example	Comments
LoginUsername	user01	Client's username
LoginPassword	password01	Client's password
SenderCompID	CLIENT01	Unique client's quote session identifier (provided by ATP)
TargetCompID	ATPFIX	Unique ATP identifier (provided by ATP)
SocketConnectHost	demo.atplatform.cn	Host address of ATP's FIX Server
SocketConnectPort	20913	Port of ATP's FIX Server
ResetOnLogon	YES	Predefined by ATP, always set to "YES"

7. Conformance Test

ATP does not have a strict set of conformance tests required for the Client FIX API connection, and therefore we advise the Client to provide a list of test scenarios they expect to encounter while in PROD.

Below are some examples of basic conformance test scenarios to be carried out between ATP and the Client (this is by no means an exhaustive list):

- login and logout
- check the behavior (reset/persistence) of Sequence numbers
- subscribe to a contract with Market Data Request to receive quote data of the contract
- subscribe multiple contracts with multiple Market Data Requests
- unsubscribe a contract with Market Data Request to stop receiving quote data of the contract
- unsubscribe multiple contracts with multiple Market Data Requests
- subscribe to a contract with incorrect contract code to receive a rejection from server
- etc.

8. Support and FAQs

For support or questions about the ATP FIX Engine, please contact the support team of ATP.

9. Appendix

9.1. ATP FIX Tag

Tag	Field Name
7	BeginSeqNo
8	BeginString
9	BodyLength
10	Checksum
16	EndSeqNo
34	MsgSeqNum
35	MsgType
36	NewSeqNo
45	RefSeqNum
49	SenderCompID
50	SenderSubID
52	SendingTime
55	Symbol
56	TargetCompID
57	TargetSubID
58	Text
95	RawDataLength
96	RawData
98	EncryptMethod
108	HeartBtInt
112	TestReqID
123	GapFillFlag
141	ResetSeqNumFlag
146	NoRelatedSym
262	MDReqID
263	SubscriptionRequestType
264	MarketDepth

265	MDUpdateType
267	NoMdEntryTypes
268	NoMDEntries
269	MDEntryType
270	MDEntryPx
271	MDEntrySize
279	MDUpdateAction
290	MDEntryPositionNo
371	RefTagID
372	RefMsgType
373	SessionRejectReason
553	Username
554	Password